A stone from Mars: Teghaza 001

CHEAT SHEET

@nova.xplains

What was found?

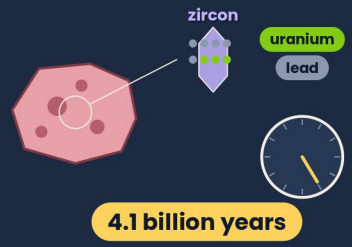


Found in 2022 on a salt plain in Mali (Sahara)

Rose-colored, about 800 g: as heavy as a pineapple

A meteorite = a space rock that survives the fall and lands

How old is it?

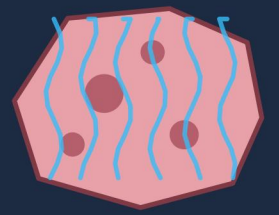


Tiny zircon crystals hold uranium that slowly turns into lead

Like a clock: measure the lead, read the age

About 4.1 billion years old · Earth is about 4.5 billion

Ancient water



Water seeped through the rock while it was still on Mars

It left chemical fingerprints in the minerals

Mars today: cold, dry desert · 4 billion years ago: liquid water

Leaving Mars

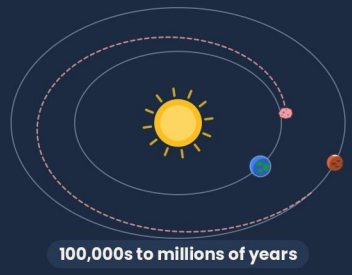


A big asteroid impact blasts rocks into space

Escape velocity on Mars: about 5 km per second

= about 18,000 km/h, around 20x faster than a jet

The long journey



The rock circles the Sun on its own path

Mars meteorites travel 100,000s to millions of years

Earth's gravity catches it · deserts are perfect for finding them

How we know: Mars!



1976: Viking landers measured the gases in Mars' air

Gas bubbles in some meteorites match exactly

Teghaza 001 has the same oxygen mix · about 435 Mars meteorites known

BONUS Quiz: To escape Mars, a rock needs about 5 kilometers per second (answer B).

Sources: Science (Liu et al., Oct 2026); NASA JPL; Meteoritical Bulletin; NASA (Viking, Mars meteorites)

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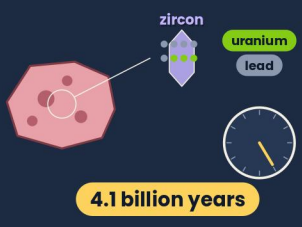


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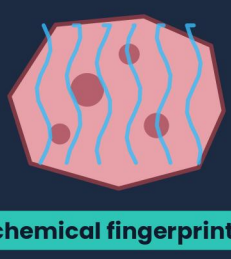


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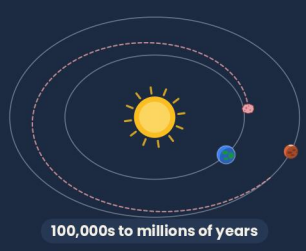


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